

Chapter 8: Experimental Predictions and Spectral Anomalies

8.1 The Operational Mandate

The BMI framework rejects the untestable nature of modern string theories. By deriving particles as harmonic nodes of a continuous background, we provide deterministic, testable predictions. The BMI project invites the experimental community to verify the geometric structure of the vacuum through high-luminosity spectral sweeps.

8.2 Mass Anomaly Predictions

Mass anomalies are non-perturbative responses to local environmental constraints, coupled to the global **Metric Equilibrium** (Δ). The predicted shift is scaled by the current cosmological epoch:

$$\delta M_n(t) = n \cdot \omega_0 \cdot [\Lambda_{tensor} \cdot f(\Delta(t)) \cdot (E^2 + c^2 B^2)]$$

This coupling means that experimental mass measurements are not just data points; they are probes of the current **Bridge Stress** (τ) and **Inter-brane Potential** (V_{gap}) of the universe.

8.3 The "Dark" Spectrum

We predict the existence of short-lived resonance states (the "Dark Spectrum") suppressed in relaxed vacuum but visible under high tension. Key targets include:

- **Sub-Mesonic Node (115.4±0.2 MeV)**: A meta-stable catch in the manifold relaxation.
- **Baryonic Cluster Split (962.8±0.5 MeV)**: A high-tension mode approaching the topological breaking limit.

8.4 The Topological Breaking Limit (Ω_{max})

The **Topological Breaking Limit** is the absolute energy threshold where nested torus geometry fails, and energy dissipates into gravitational strain. This defines the final, hard physical limit of the particle zoo, ensuring the theory remains falsifiable and bounded.